

Evolutionary Adaptation of an HP1-protein Chromodomain Integrates Chromatin and DNA Sequence Signals

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Abstract

Members of the diverse heterochromatin protein 1 (HP1) family play crucial roles in heterochromatin formation and maintenance. Despite the similar affinities of their chromodomains for di- and tri-methylated histone H3 lysine 9 (H3K9me2/3), different HP1 proteins exhibit distinct chromatin binding patterns, likely due to interactions with various specificity factors. Here, we elucidate the molecular basis of the interaction between the HP1 protein Rhino, a crucial factor of the *Drosophila* piRNA pathway, and Kipferl, a DNA sequence-specific C₂H₂ zinc finger protein and Rhino guidance factor. Through phylogenetic analyses, structure prediction, and *in vivo* genetics, we identify a single amino acid change within Rhino's chromodomain, G31D, that does not affect H3K9me2/3 binding but disrupts the interaction between Rhino and Kipferl. Flies carrying the *rhino*^{G31D} mutation phenocopy *kipferl* mutant flies, with Rhino redistributing from piRNA clusters to satellite repeats, causing pronounced changes in the ovarian piRNA profile of *rhino*^{G31D} flies. Thus, Rhino's chromodomain functions as a dual-specificity module, facilitating interactions with both a histone mark and a DNA-binding protein.

eLife assessment

This **fundamental** work has completed our understanding of the singular binding profile of the Rhino HP1 protein to chromatin, a key step in converting certain genomic regions into piRNA source loci. The evidence supporting the conclusions is **compelling**. Phylogenetic analyses, structure prediction, rigorous biochemical assays and *in vivo* genetics emphasize the importance of the Rhino chromodomain in the recognition of both a histone mark and a DNA-binding protein, and highlight the importance of a single chromodomain residue in the protein-protein interaction.

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Introduction

Transposable elements (TEs) are ubiquitous in eukaryotic genomes but pose a significant threat to genome integrity (Bourque et al., 2018). When activated and mobile, these selfish genetic elements can lead to insertional mutagenesis and ectopic recombination events, imposing significant fitness costs on their hosts. To counteract the deleterious effects of TEs, eukaryotes package TE loci into repressive heterochromatin, effectively silencing these elements and preventing their uncontrolled movement within the genome (Levin and Moran, 2011, Fedoroff, 2012). Proteins of the heterochromatin protein 1 (HP1) family play a central role in the initiation and maintenance of heterochromatin from fungi to animals (Vermaak and Malik, 2009).

The founding member of the HP1 family, *Drosophila* Su(var)2-5, acts as a strong suppressor of position effect variegation (James and Elgin, 1986, Eissenberg et al., 1990, Eissenberg et al., 1992). It binds to heterochromatic histone marks and facilitates transcriptional silencing and the compaction of chromatin through the recruitment of histone methyltransferases, histone deacetylases, and other repressive activities (Vermaak and Malik, 2009, Allshire and Madhani, 2018). Most animal genomes encode multiple HP1 homologs that share a common domain architecture. They contain an N-terminal chromodomain with specific affinity for di- and trimethylated histone H3 Lysine 9 (H3K9) peptides (Bannister et al., 2001, Lachner et al., 2001), an unstructured central hinge region of variable length involved in nonspecific nucleic acid interactions (Keller et al., 2012), and a C-terminal chromo shadow domain (Aasland and Stewart, 1995). While resembling the chromodomain fold, the chromo shadow domain does not bind histone tails. Instead, it forms a dimerization interface with the chromo shadow domain of another HP1 protein, creating a binding groove for proteins containing a PxV/LxL consensus motif (Smothers and Henikoff, 2000).

The number of HP1 family members varies between species. For instance, mice and humans encode three HP1 family proteins (HP1 α , HP1 β , HP1 γ), whereas *Drosophila melanogaster* encodes five different members: the ubiquitously expressed HP1a/Su(var)2-5, HP1b, and HP1c proteins, and the germline-specific HP1d/Rhino (ovary and testis) and HP1e (testis) proteins (Vermaak and Malik, 2009, Levine et al., 2012). Despite having similar affinities for H3K9me2/3 reported from *in vitro* experiments, the *Drosophila* HP1 proteins have distinct biological functions and chromatin-binding patterns (Yu et al., 2015, Lee et al., 2019, Baumgartner et al., 2022). For example, while Su(var)2-5 binds all H3K9-methylated loci genome-wide, the germline-specific Rhino is enriched only at specific heterochromatic loci from where the non-coding precursors of PIWI-interacting RNAs (piRNAs) are transcribed (Vermaak et al., 2005, Klattenhoff et al., 2009, Mohn et al., 2014, Zhang et al., 2014). These so-called piRNA clusters are rich in repetitive sequences and serve as heritable sequence storage units that confer specificity to the piRNA pathway, a small RNA-based TE silencing system in animal gonads (Brennecke et al., 2007, Czech et al., 2018, Ozata et al., 2018). At the molecular level, Rhino facilitates the productive expression of heterochromatic piRNA clusters by recruiting specific effector proteins that stimulate transcription initiation, elongation, and nuclear export of the resulting non-coding piRNA precursors (Klattenhoff et al., 2009, Mohn et al., 2014, Zhang et al., 2014, Chen et al., 2016, Andersen et al., 2017, ElMaghraby et al., 2019, Kneuss et al., 2019). This makes Rhino a remarkably specialized HP1 protein that mediates activating, rather than repressive, chromatin identity. The precise regulation of Rhino's chromatin-binding profile is therefore of great importance.

The zinc finger protein Kipferl, one of about ninety ZAD zinc finger proteins in *Drosophila*, acts as a critical guidance factor for Rhino in ovaries (Baumgartner et al., 2022). Kipferl binds to chromatin at genomic sites enriched in GRGGN motifs, presumably through a direct interaction between its C₂H₂ zinc finger arrays and DNA. When genomic Kipferl binding sites are located

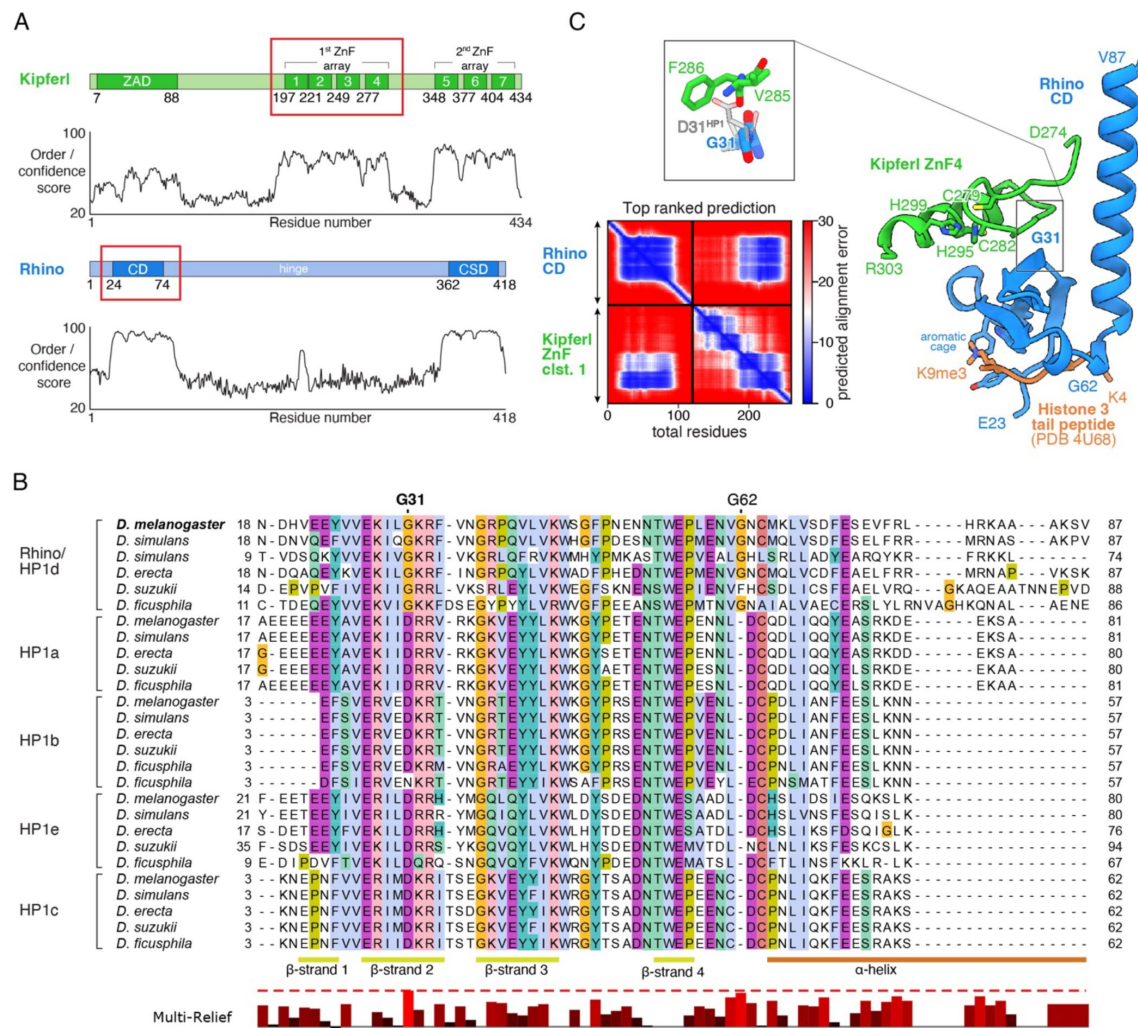


Figure 1

Structure prediction and phylogenetic analyses point to a Rhino-specific residue involved in binding Kipferl.

(A) Domain organization of Kipferl and Rhino, with the AlphaFold pLDDT score plotted as a measure of order or disorder alongside. Red boxes indicate the smallest interacting fragments identified by yeast two-hybrid experiments by Baumgartner et al. (Baumgartner et al., 2022). ZAD, Zinc finger associated domain; ZnF, Zinc finger; CD, chromodomain; CSD, chromo shadow domain (B) Multiple sequence alignment of HP1 family proteins in five selected species harboring an unequivocally identified Kipferl homolog (see Figure 1 – figure supplement 2). Rhino-specific amino acid residues are indicated. Protein accessions and identifiers are documented in Supplementary File 1. Multi-Relief representation indicates residues that differ significantly in Rhino homologs versus other HP1 variant proteins. Note that two Rhino paralogs are identified in *D. simulans* (see Supplementary File 1 for accessions). (C) PAE plot for the top ranked AlphaFold2 Multimer prediction of the Rhino chromodomain with the Kipferl ZnF cluster 1 (left) and structure of the complex in cartoon representation (Rhino in blue; Kipferl in green), together with the H3K9me3 peptide (orange) as observed in a Rhino–H3K9me3 crystal structure (PDB ID 4U68). Key residues of Rhino’s aromatic cage and H3K9me3, as well as of Kipferl’s C₂H₂ ZnF4 are shown in sticks representation. Only the interacting ZnF4 is shown. Depicted in the inset are Rhino G31 and HP1 D31, with HP1 (PDB ID 6MHA) superimposed on Rhino chromodomain residues 26–57 (RMSD = 0.55 Å), together with Kipferl V285 and F286, illustrating that D31 would lead to steric clashes with Kipferl.

The Rhino^{G31D} chromodomain retains H3K9me3 binding *in vitro*

Rhino's *in vivo* function depends critically on its ability to bind H3K9me2/3 via its chromodomain (Yu et al., 2015 [\[1\]](#)). In addition, dimerization of the Rhino chromodomain has been suggested to be important for its function (Yu et al., 2015 [\[1\]](#)). To determine whether the Rhino G31D mutation affects either of these functions, we analyzed a panel of recombinantly expressed Rhino chromodomains. This panel included the wildtype construct, two putative Kipferl-binding mutants (G31A and G31D), and control mutants that impair H3K9me2/3 binding (mutations of the aromatic cage residues Y24A, W45A, or F48A) or putative dimerization (F34A/F76A double mutant) (Yu et al., 2015 [\[1\]](#)).

We used analytical size-exclusion chromatography with inline multi-angle light scattering (SEC-MALS) to assess the oligomeric state of the different Rhino chromodomain constructs. Our data confirmed differences in elution volume among the different mutant constructs (Yu et al., 2015 [\[1\]](#)), but these differences did not correspond to significant changes in their in-solution molecular weight, indicating that the oligomeric state remained consistent across all constructs tested (Figure 2A [\[1\]](#); Figure 2 – figure supplement 1 [\[1\]](#)). We conclude that the isolated wildtype Rhino chromodomain, along with the G31D or G31A variants, are monomeric in solution, as has been shown for other HP1 homologs (Jacobs et al., 2001 [\[2\]](#), Brasher et al., 2000 [\[3\]](#)). To further investigate whether the G31D mutation causes any unwanted structural changes in the Rhino chromodomain, we performed circular dichroism spectroscopy. All tested mutant constructs exhibited similar secondary structure compositions compared to the wildtype construct (Figure 2 – figure supplement 2 [\[1\]](#)).

Having established that the two G31 mutant chromodomains do not exhibit altered protein folding or oligomeric state, we tested both constructs for their ability to bind H3K9me3 peptides alongside wildtype and aromatic cage mutant (F48A) controls. Consistent with previous observations (Yu et al., 2015 [\[1\]](#), Le Thomas et al., 2014 [\[4\]](#)), isothermal titration calorimetry (ITC) experiments using synthetic H3K9me3 peptides revealed an affinity of $30.9 \pm 3.0 \mu\text{M}$ for the wildtype domain and no measurable affinity for the F48A mutant (Figure 2B [\[1\]](#)). Despite slight changes in the thermodynamic binding parameters, both the G31A and G31D mutants showed affinities comparable to the wildtype constructs with $43.5 \pm 8.6 \mu\text{M}$ and $31.1 \pm 3.2 \mu\text{M}$, respectively. Thus, the Rhino^{G31D} chromodomain behaves similarly to the wildtype domain in terms of oligomeric state, folding, and ability to bind H3K9me3 peptides *in vitro*.

The *rhino*^{G31D} mutant uncouples Rhino from Kipferl

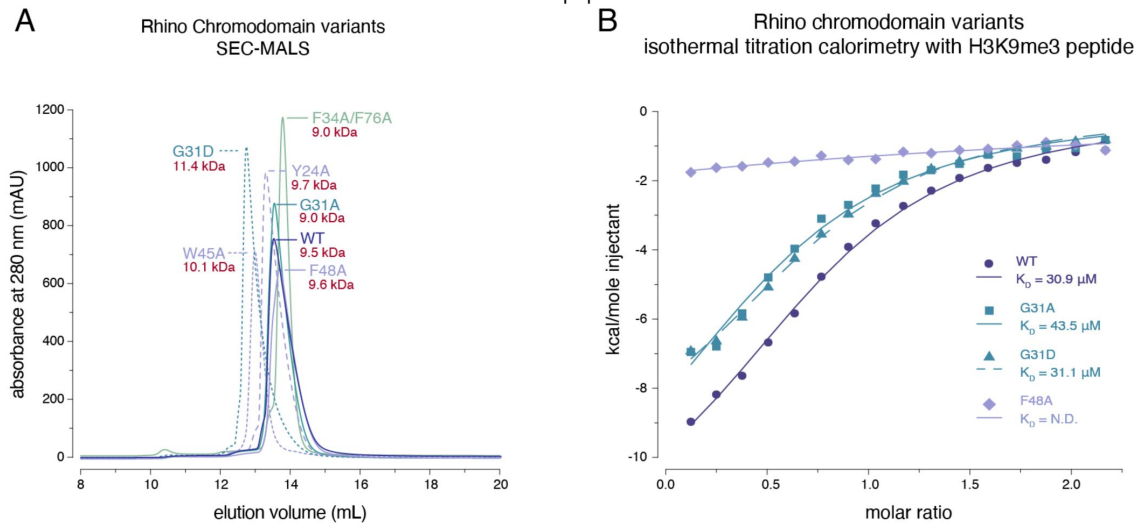
To explore the importance of G31 for Rhino function *in vivo*, we engineered a single point mutation within the endogenous *rhino* locus, converting G31 to the aspartic acid residue typically present in all other HP1 proteins (*rhino*^{G31D}). In *kipferl* mutant females, Rhino fails to localize to the majority of its genomic binding sites, resulting in diminished piRNA levels and compromised fertility (Baumgartner et al., 2022 [\[5\]](#)). Homozygous females carrying the *rhino*^{G31D} allele were viable but exhibited severely reduced fertility: Although the egg-laying rate of *rhino*^{G31D} females was comparable to that of control flies, the hatching rate of laid eggs dropped to $21 \pm 9\%$ (Figure 3A [\[5\]](#)). While this decline in fertility was less severe compared to the complete sterility observed in *rhino* null mutants, it closely mirrored the impaired fertility of *kipferl* null mutants, which was in the range of 15 to 40% (Baumgartner et al., 2022 [\[5\]](#)), providing a first indication that the G31D mutation may specifically affect the Rhino–Kipferl interaction.

To gain deeper insights into the Rhino–Kipferl interaction in *rhino*^{G31D} mutants, we examined changes to the pronounced colocalization of Kipferl and Rhino at discrete nuclear foci – corresponding to piRNA source loci – observed in wild-type nurse cells (Baumgartner et al., 2022 [\[5\]](#)). Using immunofluorescence imaging, we observed a complete absence of colocalization

Figure 2

Rhino G31 point mutations do not affect Rhino's ability to bind H3K9me3.

(A) Line graph summarizing SEC-MALS results for the examined Rhino chromodomain constructs. The in solution molecular weight is indicated for each construct. (B) Isothermal titration calorimetry results showing the binding of indicated Rhino chromodomain constructs to the H3K9me3-modified histone tail peptide.

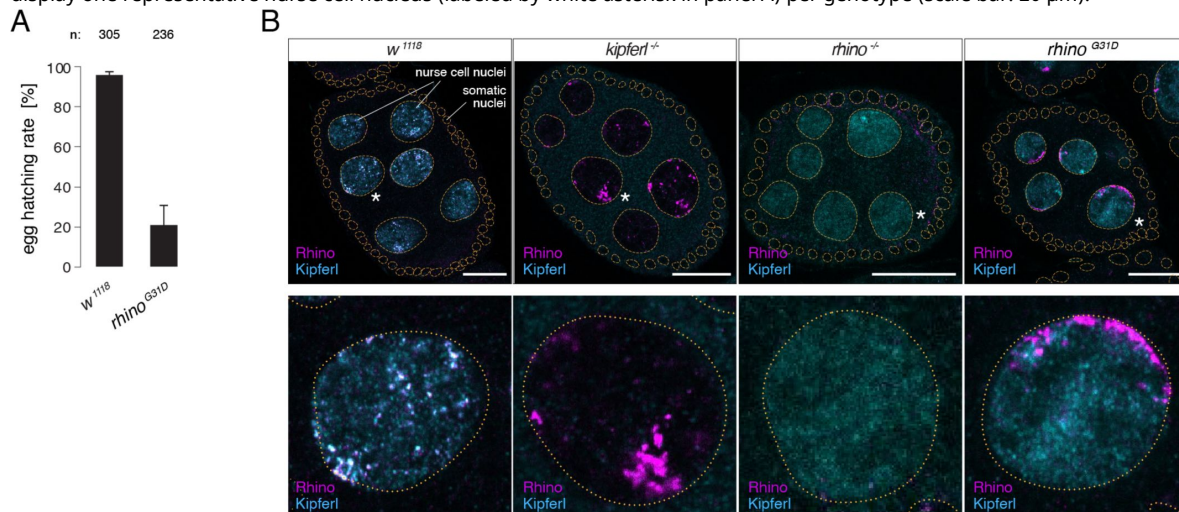


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Figure 3

The *rhino*^{G31D} point mutation recapitulates the phenotypes for Rhino and Kipferl in each other's null mutant background.

(A) Bar graph depicting female fertility as egg hatching rate in percent of laid eggs for indicated genotypes. (B) Confocal images showing immunofluorescence signal for Kipferl and Rhino in egg chambers of indicated genotypes. Zoomed images display one representative nurse cell nucleus (labeled by white asterisk in panel A) per genotype (scale bar: 20 μm).

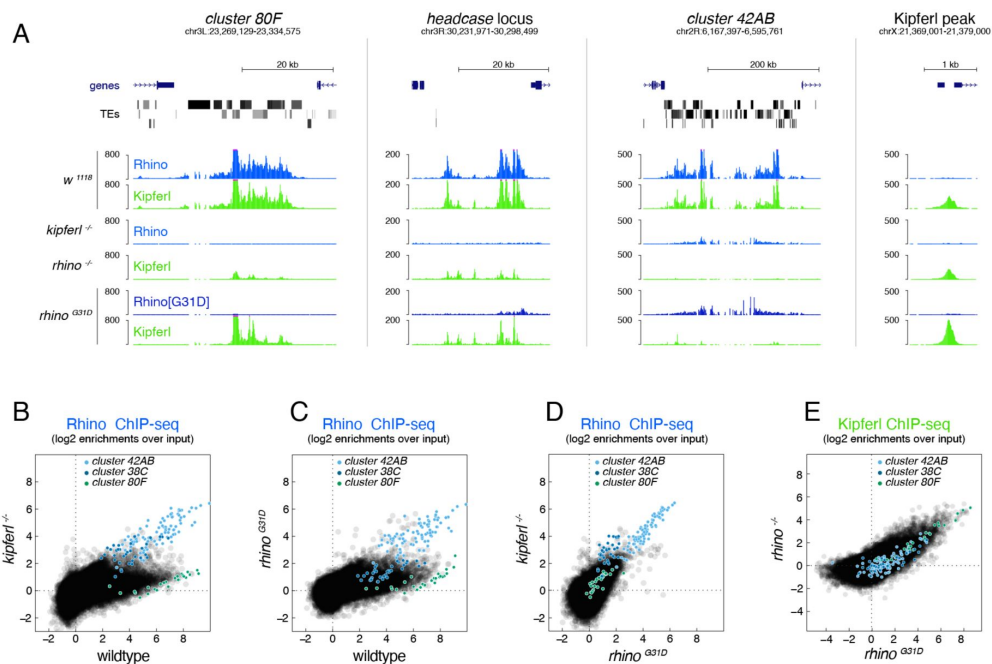


between Rhino and Kipferl in *rhino*^{G31D} mutants (**Figure 3B**). Kipferl localized diffusely in the nucleus with only a few foci, mirroring its distribution in *rhino* null mutants. Rhino^{G31D} was not enriched within these Kipferl foci; instead, it accumulated in prominent structures near the nuclear envelope, resembling the Rhino accumulations found in *kipferl* null mutants (Baumgartner et al., 2022).

To determine the chromatin binding patterns of Rhino and Kipferl in ovaries of *rhino*^{G31D} mutant flies, we performed chromatin immunoprecipitation followed by sequencing (ChIP-seq). In wild-type ovaries, Rhino and Kipferl co-occupy hundreds of heterochromatic domains, displaying nearly identical enrichment patterns (**Figure 4A**). In addition, Kipferl, but not Rhino, binds to specific sites in euchromatin (Kipferl-only sites) that lack H3K9me2/3 marks but are enriched in GRGGN motifs, Kipferl's presumed DNA binding motif. To account for the heterogeneous size of genomic Rhino/Kipferl domains, we analyzed their binding profiles by quantifying genome-unique ChIP-seq reads mapped to non-overlapping genomic 1-kilobase tiles. In *kipferl* mutants, Rhino is lost from most of its genomic binding sites, with retained Rhino binding primarily corresponding to piRNA clusters 38C and 42AB (**Figure 4A, B**). Conversely, in *rhino* mutants, Kipferl binding persists at euchromatic Kipferl-only sites but is strongly reduced at loci that are co-occupied by Kipferl and Rhino in wildtype: at sites where Rhino binding is Kipferl-dependent, Kipferl binding is reduced to more defined, narrow peaks. At Kipferl-independent loci on the other hand (e.g., piRNA clusters 38C and 42AB), Kipferl binding is almost completely lost in *rhino* mutants (**Figure 4A**). ChIP-seq experiments in *rhino*^{G31D} mutant ovaries revealed a chromatin occupancy for Rhino^{G31D} that was almost indistinguishable from that of wild-type Rhino in *kipferl* mutants (**Figure 4C, D**). This similarity extended to Kipferl-independent loci (e.g., piRNA clusters 38C and 42AB), where the altered chromatin occupancy of Rhino in *kipferl* mutants was mirrored by Rhino^{G31D} (**Figure 4A**). At the same time, the chromatin binding profile of Kipferl in *rhino*^{G31D} mutants strongly resembled that observed in *rhino* null-mutants genome-wide (**Figure 4A, E**). Taken together, the mutation of a single Rhino-specific chromodomain residue to its ancestral state results in the functional uncoupling of Rhino and Kipferl at the molecular level.

Rhino^{G31D} is functional at Kipferl-independent piRNA source loci

To assess the impact of the Rhino^{G31D} point mutation on Rhino's overall functionality, we analyzed Kipferl-independent but Rhino-dependent piRNA source loci. In *kipferl* mutant ovaries, Rhino is sequestered to large DNA satellite arrays, resulting in greatly increased transcription and piRNA production at the *Responder* and 1.688 g/cm3 family satellites (Baumgartner et al., 2022). In *rhino*^{G31D} mutants, RNA fluorescent in situ hybridization (FISH) experiments showed that transcription of the *Rsp* and 1.688 g/cm3 satellites was also strongly elevated, leading to elongated structures at the nuclear envelope, reminiscent of the phenotype observed in *kipferl* mutant nurse cell nuclei (**Figure 5A**). Consistent with this elevated transcription, Rhino^{G31D} was enriched at satellite consensus sequences as determined by ChIP-seq, while it was reduced at most transposon sequences (**Figure 5B, C**). These findings extended to piRNA levels: piRNAs originating from *Rsp* and 1.688 g/cm3 satellites were substantially increased (**Figure 5D**), while piRNAs were reduced at Kipferl-dependent piRNA clusters (e.g. cluster 80F), but not at Kipferl-independent piRNA clusters like 38C and 42AB (**Figure 5E**). Similarly, the levels of piRNAs mapping to transposon consensus sequences showed similar behaviors in *rhino*^{G31D} mutants as observed in *kipferl* mutants (**Figure 5 – figure supplement 1A**). This provides further confirmation that the Rhino^{G31D} mutation faithfully phenocopies a *kipferl* null-mutant, indicating that Rhino^{G31D} remains fully functional at Kipferl-independent loci. The altered piRNA levels observed in *kipferl* mutant ovaries result in the de-repression of a handful of transposable elements (Baumgartner et al., 2022). Based on RNA FISH experiments, the same transposons were also de-repressed in *rhino*^{G31D} females, with the levels of upregulation resembling those in *kipferl* mutants rather than *rhino* mutants (**Figure 5 – figure supplement 2**), further suggesting a specific requirement of G31 for Kipferl-dependent functions of Rhino.



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Figure 4

The Rhino^{G31D} point mutation uncouples Rhino and Kipferl on chromatin.

(A) UCSC genome browser screenshots depicting the ChIP-seq signal for Rhino and Kipferl at diverse Rhino domains in ovaries of the indicated genotypes (signal shown as coverage per million sequenced reads for one representative replicate). (B-E) Scatter plot of genomic 1-kb tiles contrasting average log2-fold ChIP-seq enrichment for Rhino (B-D) or Kipferl (E) in ovaries of the indicated genotypes (values displayed represent the average of two to three replicate experiments).

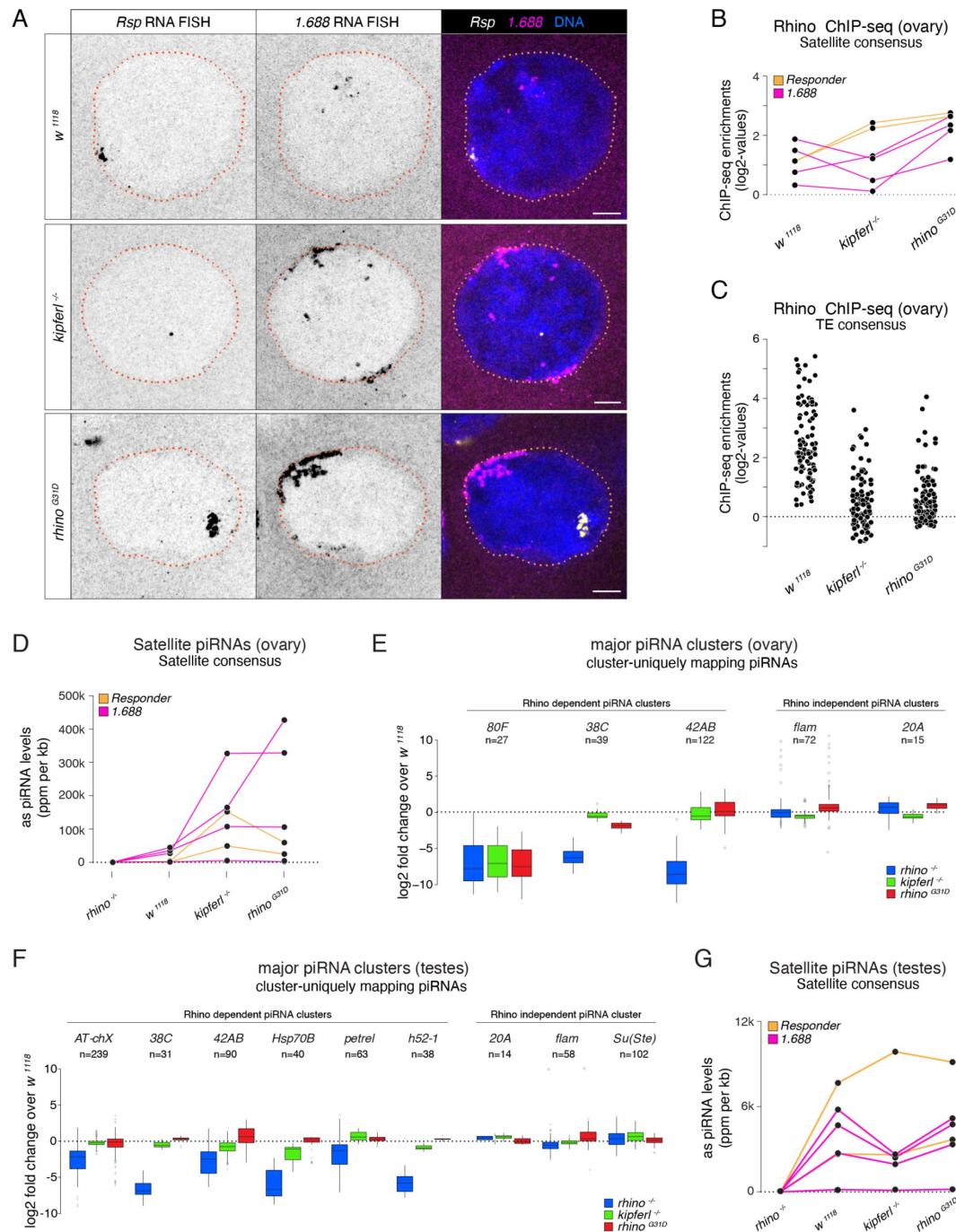


Figure 5

Kipferl-independent functions of Rhino are not affected by the G31D mutation.

(A) Confocal images showing *Rsp* and 1.688 *g/cm3* Satellite RNA FISH signal in nurse cells of indicated genotypes (scale bar: 5 μ m). (B, C) Jitter plots depicting the log₂-fold enrichments for Rhino ChIP-seq on consensus sequences of Satellites (B) or Rhino-dependent transposons (C) in ovaries with indicated genetic backgrounds. (D, G) Jitter plots depicting the length-normalized antisense piRNA counts on Satellite consensus sequences derived from ovaries (D) or testes (G) of indicated genetic backgrounds. (E, F) Box plots depicting the log₂ fold change of piRNA counts (compared to *w¹¹¹⁸* control) per 1kb tile for major piRNA clusters in ovaries (E) or testes (F) of the indicated genotypes. The number of tiles per piRNA cluster is indicated (n).

Rhino also plays a role in specifying piRNA source loci in the male germline, where Kipferl is not expressed (Chen et al., 2021 [↗](#), Chen and Aravin, 2023 [↗](#), Baumgartner et al., 2022 [↗](#)). The piRNA source loci in testes only partially overlap with those of ovaries and are dynamically regulated during spermatogenesis, suggesting Kipferl-independent mechanisms for Rhino recruitment to chromatin. To assess a potential impact of the G31D mutation on Rhino function in males, we sequenced testes small RNAs from a panel of different genetic mutants. Comparing piRNAs from *rhino* mutant testes to wildtype controls confirmed the expected loss of piRNA production specifically from dual-strand piRNA source loci, whereas piRNA levels from the same loci remained unchanged in *kipferl* or *rhino*^{G31D} mutants (**Figure 5F** [↗](#)). Consistent with this, the levels of transposon-mapping piRNAs also remained unaltered in *kipferl* or *rhino*^{G31D} mutants (**Figure 5 – figure supplement 1B** [↗](#)). The lack of a piRNA phenotype in testes further extended to the *Rsp* and *1.688 g/cm3* satellite loci, which produce Rhino-dependent piRNAs also in the male germline (**Figure 5G** [↗](#)). Taken together, the G31D mutation, while completely uncoupling Rhino from Kipferl, does not impede Rhino function at Kipferl-independent sites in either ovaries or testes.

Conclusion

In this study we elucidate the intricate interplay between the DNA sequence-specific zinc finger protein Kipferl, and the chromodomain of the HP1 variant Rhino. Our findings underscore the critical role of Kipferl in orchestrating Rhino's chromatin-binding dynamics and subsequent piRNA production. Specifically, we show that a single amino acid alteration within Rhino's chromodomain, reverting it to its ancestral state (G31D), disrupts Kipferl's ability to target Rhino to chromatin. Notably, the G31 residue in Rhino is highly conserved among *Drosophilids*, even in species that lack a clearly identifiable Kipferl ortholog. This may indicate that other proteins use a mechanism similar to Kipferl to define Rhino's chromatin occupancy in more distantly related *Drosophila* species. Our data also show that the *rhino*^{G31D} mutation does not affect the chromatin binding or the function of Rhino at Kipferl-independent piRNA source loci in ovaries and testes, suggesting the existence of other, G31-independent mechanisms for recruitment of Rhino to chromatin. Whether these alternative mechanisms act in a similar way to the one described here, utilizing zinc finger proteins and interactions with the Rhino chromodomain, remains an open question. An important issue for future investigation, currently hampered by the challenges of obtaining soluble recombinant Kipferl protein, will be to determine the precise three-dimensional arrangement of the Kipferl-Rhino complex together with Kipferl motif-containing DNA and H3K9-methylated nucleosomes, considering that Kipferl and Rhino are both likely to form homodimers via their N-terminal ZAD domain and C-terminal chromo shadow domain, respectively.

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Data and material availability

Sequencing data sets have been deposited to the NCBI GEO archive (GSE244196). Previously published data sets analyzed in this study are listed in Supplementary File 2. All fly strains generated for this study are available via the VDRC (<http://stockcenter.vdrc.at/control/main> .

Declaration of interests

The authors declare no competing interests.

Supplementary files

Supplementary File 1

Supplementary File 2

Figure 1-figure supplement 1 

Figure 1-figure supplement 2 

Figure 1-figure supplement 3 

Figure 1-figure supplement 4 

Figure 2-figure supplement 1 

Figure 2-figure supplement 2 

Figure 5-figure supplement 1 

Figure 5-figure supplement 2 

Key Resource Table

Materials & methods

Fly strains and husbandry

All fly stocks were maintained at 25°C with 12h dark/light cycles. Fly strains used in this study are listed in the Key Resource Table. For ovary dissections, flies were aged for 2–6 days and held in cages with apple juice plates and fresh yeast paste for two days. Flies harboring the *rhino*^{G31D} point mutation were generated from isogenised *w*¹¹¹⁸ embryos by co-injecting the pDCC6b plasmid (Gokcezade et al., 2014 [DOI](#)) expressing a gRNA (TATGTAGTGGAGAAAATCTT) with an HDR donor oligo (GGTCGATGCACCGCTAAATGATCATGTCTGAAGAATATGTAGTGGAGAAAATCcTgGatAAACGTTTGTAAATGGGCGTCCCCAGTTCTGGTGAAGTGGAGCGGTTTTCCG; IDT).

Phylogenetic analyses

Kipferl and related zinc finger associated domain-containing (zf-AD) proteins were collected with NCBI BLAST searches using *Drosophila melanogaster* Kipferl zf-AD (region 5-95) in the NCBI non-redundant protein or the UniProt reference proteomes databases (Altschul et al., 1997 [DOI](#), UniProt, 2021 [DOI](#), Coordinators, 2018 [DOI](#)) applying significant E-value thresholds ($1e^{-5}$). Selected proteins, covering the zf-AD over the complete length, were aligned with MAFFT (v7.505, -linsi method) (Kato and Toh, 2008 [DOI](#)), and the zf-AD region extracted with Jalview (Waterhouse et al., 2009 [DOI](#)). A maximum likelihood phylogenetic tree was calculated with IQ-TREE 2 (v2.2.0) (Minh et al., 2020 [DOI](#)), with standard model selection using ModelFinder (Kalyaanamoorthy et al., 2017 [DOI](#)) and ultrafast bootstrap (UFBoot2) support values (Hoang et al., 2018 [DOI](#)). The tree was visualized in iTOL (v6) (Letunic and Bork, 2021 [DOI](#)). Branches that are supported by an ultrafast bootstrap (UFBoot) value $\geq 95\%$ are indicated by a grey dot. Branch lengths represent the inferred number of amino acid substitutions per site, and branch labels are composed of gene name (if available), genus, species, and accession number. A similar approach was performed to collect Rhino and HP1 sequences. Full length *D. melanogaster* HP1-like sequences were used as query for blast searches applying highly significant E-value thresholds ($1e^{-10}$). Only sequences covering both chromodomain (CD) and chromo shadow domain (CSD) were considered for further analysis. The alignment was condensed by removing all columns covering less than 70% of the sequences and a maximum likelihood phylogenetic tree was inferred. To search for residues in Rhino that are distinct from all other HP1-like families, we focused on 17 *Drosophila* species where Kipferl could be detected and extracted 104 protein sequences. In the resulting alignment, sub-family specific residues were detected with the multi-relief method (<https://www.ibi.vu.nl/programs/shmrwww/> [DOI](#)) (Brandt et al., 2010 [DOI](#)).

AlphaFold predictions

AlphaFold2-Multimer (Jumper et al. 2021 [DOI](#), Evans et al. 2021) was used to predict protein-protein interactions on a local GPU cluster with a script using MMseqs2 (Steinegger and Soding, 2017 [DOI](#)) (git@92deb92) for local MSA creation and Colabfold (Mirdita et al., 2022 [DOI](#)) (git@7227d4c) for structure prediction. Protein structures were analyzed using ChimeraX (Pettersen et al., 2021 [DOI](#)).

Expression and purification of the Rhino chromodomain

His₆-SUMO-RhinoCD constructs (spanning Rhino residues 20-90 in the vector pET-28) were transformed into the *E. coli* strain BL21-CodonPlus (DE3)-RIPL (Agilent) for large-scale expression using standard methods. Briefly, cultures were grown in Terrific Broth media supplemented with appropriate antibiotic(s) at 37°C to a culture density of approximately OD_{λ=600 nm} of 1.2. Cultures were then cooled in an ice water bath for 15 minutes followed by induction of protein expression with 0.5 mM IPTG. Induction proceeded overnight at 16°C with shaking at 220 rpm. Cells were harvested by centrifugation at 4000g for 30 minutes at 4°C. For Ni-NTA purification, cell pellets

were resuspended in 20 mL lysis buffer (50 mM sodium phosphate, pH 8.0, 50 mM NaCl, 10 mM imidazole, 10 µg/mL DNase I, and protease inhibitors) per liter culture. The resuspended cells were lysed by sonication and the lysate was then clarified by ultracentrifugation at roughly 140,000g for 30 minutes. The soluble supernatant was taken for affinity purification via Ni-NTA column (1.5 mL of beads per liter culture), pre-equilibrated with lysis buffer. Beads were washed with 10 column volumes of wash buffer (50 mM sodium phosphate, pH 8.0, 200 mM NaCl, 10 mM imidazole) followed by elution of the target protein in 50 mM sodium phosphate, pH 8.0, 100 mM NaCl, 150 mM imidazole. To remove the affinity tag, Ulp1 protease was added in a 1:10 mass ratio (protease:RhinoCD) and incubated overnight at 4°C. 1 mM EDTA and 5 mM DTT (final concentrations) were added to limit degradation and enhance tag cleavage, respectively. The protein was further purified using tandem ion exchange chromatography with HiTrap Q HP and HiTrap SP HP columns (Cytiva/GE Healthcare Life Sciences). Digested protein was first diluted three-fold with low salt buffer (20 mM Tris, pH 7.5, 1 mM DTT) then applied to the HiTrap Q column. The flowthrough was collected and purified using the HiTrap SP column. The target protein was eluted using a 0-1 M NaCl gradient in 20 mM Tris, pH 7.5, and 1 mM DTT over approximately 60 mL. Peak fractions were assessed by SDS-PAGE then selected and pooled for further purification. Pooled fractions were concentrated and further purified by gel filtration chromatography using a Superdex75 column equilibrated with 20 mM Tris, pH 7.5, 150 mM NaCl, 1 mM DTT. Depending on the total yield, either a Superdex75increase 10/300 column or a Superdex75 HiLoad 16/600 column (Cytiva/GE Healthcare Life Sciences) was used. Peak fractions were assessed by SDS-PAGE. Fractions with highly purified protein were concentrated, then stored at 4°C. For long-term storage the protein was flash frozen in liquid nitrogen then kept at -80°C. Typical yields were 1-10 mg of purified protein (>98% pure as assessed by SDS-PAGE) per liter culture.

Size exclusion chromatography with inline multiangle light scattering (SEC-MALS)

Multiangle light scattering was used to determine the oligomeric state of the purified proteins. Roughly 400 µg of purified protein (100 µL at 4 mg/mL) was taken for in-line size exclusion chromatography on a Superdex75increase 10/300 GL column (monitored at 280 nm) followed by light scattering analysis. Chromatography was performed in a buffer of 20 mM Tris, pH 7.5, 150 mM NaCl. MALS was measured with a Wyatt Dawn Heleos-II and processed using the included software (ASTRA Version 5.3.4). Bovine Serum Albumin (BSA) was used as calibration control.

Circular dichroism

Circular dichroism was used to assess the folding of the various Rhino chromodomain constructs. Prior to data collection, proteins were exchanged into 10 mM sodium phosphate, pH 7.5, 0.15 M NaF using Zeba 7 kDa spin desalting columns (ThermoFisher Scientific) then diluted to approximately 50 µM in the desalting buffer. Samples were measured in a 0.2 mm path length demountable quartz cuvette (Hellma) and data were acquired using a Chirascan V100 Spectrometer (Precision Biomolecular Characterization Facility, Columbia University). Spectra were collected at 22°C with a data pitch of 1 nm and scan speed of 1 nm/s. Data shown are the average of three scans after buffer subtraction and presented in units of mean residue ellipticity (degrees·cm²·dmol⁻¹·residue⁻¹). Fitting was performed by DichroWeb (Miles et al., 2022 [DOI](#)) using the CONTIN-LL method (Provencher and Glockner, 1981 [DOI](#)) with reference set 3. All fits had an NRMSD of 0.1 or less.

Isothermal Titration Calorimetry (ITC)

Approximately 500 µL of each construct was dialyzed (3.5 kDa molecular weight cutoff) into 20 mM Tris, pH 8.0, 25 mM NaCl, and 2 mM βME overnight at 4°C. The protein concentration was then determined by absorbance at 280 nm after which the protein was diluted to 100 µM in dialysis buffer. H3K9me3 peptide (KQTAR-K[me3]-STGGK) was purchased from AnaSpec, Inc. and

resuspended at approximately 1 mM in dialysis buffer. Calorimetry was conducted using a MicroCal iTC200 at 20°C with stirring at 750 rpm with a reference power of 11 µcal/sec. Sixteen 2.5 µL injections were performed with an injection spacing of 120 seconds. Binding curves were analyzed using the included Origin 7 SR4 (version 7.0552 (B552)) software.

RNA Fluorescence In Situ Hybridization

RNA FISH for *Rsp* and 1.688 g/cm³ Satellites was performed using an in-house labelled probe set composed of 48 oligos or a single fluorescent oligo, respectively (Wei et al., 2021 [\[1\]](#), Gaspar et al., 2017 [\[2\]](#)). RNA FISH for *HMS-Beagle*, *Max*, *diver*, and *3S18* transposons was performed using Stellaris probes (Biosearch Technologies). Probe sequences are listed in (Baumgartner et al., 2022 [\[3\]](#)). Briefly, 5 pairs of ovaries were dissected into ice-cold PBS, fixed at room temperature for 20 min (4% formaldehyde, 0.3% Triton X-100 in PBS), washed 3 times for 5 min at RT (PBS containing 0.3% Triton X-100) followed by incubation at 4°C overnight in 70% EtOH for full permeabilization. Ovaries were rehydrated for 5 min in wash buffer (10% formamide in 2x SSC) prior to hybridization, which was done in 50 µL hybridization buffer (100 mg/mL dextran sulfate and 10% formamide in 2x SSC) overnight at 37°C using 0.5 µL *Rsp* FISH probe per sample and a final concentration of 100 nM for the 1.688 g/cm³ FISH oligo. Samples were rinsed twice in wash buffer and washed in wash buffer twice for 30 min at 37°C. Ovaries were counterstained for DNA (DAPI 1:5000 in 2x SSC) for 5 min at RT followed by 2 washes for 5 min with 2x SSC. Ovaries were mounted on microscopy slides using DAKO mounting medium (Agilent) and equalized at RT for at least 24 h before imaging on a Zeiss LSM 880 inverted Airyscan microscope. Images are shown as Z-stack across a maximum of 2 µm.

Immunofluorescence staining of ovaries

5-10 ovary pairs were dissected into ice cold PBS before fixation (4% formaldehyde, 0.3% Triton X-100, 1x PBS) for 20 min at room temperature with rotation. Fixed ovaries were washed 3 times for 5 min each in PBX (0.3% Triton X-100, 1x PBS) and blocked with BBX (0.1% BSA, 0.3% Triton X-100, 1x PBS) for 30 min at room temperature with rotation. Incubation with primary antibody was performed at 4°C overnight with antibodies diluted in BBX. After three 5-min washes in PBX, ovaries were incubated overnight at 4°C with fluorophore-coupled secondary antibodies, washed three times in PBX with DAPI in the first wash (1:50,000 dilution). The final wash buffer was carefully removed before addition of DAKO mounting medium. The samples were imaged on a Zeiss LSM 880 confocal-microscope and image processing was done using FIJI/ImageJ (Schindelin et al., 2012 [\[4\]](#)). Images are shown as Z-stack projection across a maximum of 2 µm. All relevant antibodies and dilutions are listed in the Key Resource Table.

Scoring of embryo hatching rates

To determine female fertility, 10 females were collected as virgins and aged for 2-3 days with w¹¹¹⁸ males. The hatching rate of eggs laid on apple juice plates within 4-7 hours was determined 30 h after collection (25°C) as the percentage of hatched eggs out of the total. Only plates with more than 50 eggs were included in the analysis. Wild type females were included as a control.

Definition and curation of 1 kb genomic windows

Non-overlapping 1-kb tiles were generated based on the four assembled chromosomes of the *Drosophila melanogaster* genome (dm6 assembly) and intersected with genomic piRNA cluster coordinates for annotation. Tiles with a mappability of less than 25%, as determined by intersection with genomic blocks of continuous mappability using BEDTools coverage, were excluded from all analyses (2,761 1-kb tiles). In addition, tiles with more than a threefold deviation from the median values for representative input libraries used in (Baumgartner et al., 2022 [\[3\]](#)) (18,268 1-kb tiles) or tiles with strong residual Rhino or Kipferl signal in ChIP-seq libraries prepared from the respective knockout ovaries (20 and 495 tiles, respectively) were removed.

ChIP-Seq

ChIP was performed as described previously (Lee et al., 2006 [DOI](#)). In brief, 150 μ L of ovaries were dissected into ice-cold PBS, followed by crosslinking with 1.8% formaldehyde in PBS for 10 min at room temperature, quenching with glycine, and rinsing with PBS. Samples were flash frozen in liquid nitrogen after removing all PBS. Frozen ovaries were disrupted in PBS using a Dounce homogenizer (tight) and centrifuged at low speed. The pellet was resuspended in lysis buffer. Samples were sonicated (Bioruptor) to obtain DNA fragment sizes of 200–800 bp. Samples were incubated with specific antibodies overnight at 4°C in 350–700 μ L total volume using 1/4 to 1/3 of chromatin per ChIP (antibodies are listed in Key Resource Table). 40 μ L Dynabeads (equal mixture of Protein G and A, Invitrogen) were then added and incubated for 1 h at 4°C for immunoprecipitation. Following multiple washes, immunoprecipitated protein-DNA complexes were eluted with 1% SDS. Treatment with RNase-A, decrosslinking overnight at 65°C, and proteinase K treatment were performed before clean-up using ChIP DNA Clean & Concentrator columns (Zymo Research). Barcoded libraries were prepared using the NEBNext Ultra II DNA Library Prep Kit for Illumina (NEB) according to manufacturer's instructions and sequenced on a NovaSeqSP instrument (Illumina).

Small RNA-Seq

Small RNA cloning was performed as described in (Grentzinger et al., 2020 [DOI](#)). In brief, ovaries or testes were lysed and Argonaute-sRNA complexes were isolated using TraPR ion exchange spin columns. sRNAs were subsequently purified using acidic phenol. 3' adaptors containing 6 random nucleotides plus a 5 nt barcode on their 5' end and 5' adaptors containing 4 random nucleotides at their 3' end were subsequently ligated to the small RNAs before reverse transcription, PCR amplification, and sequencing on an Illumina NovaSeqSP instrument.

Computational Analysis

ChIP-Seq Analysis

ChIP-seq reads were trimmed to remove the adaptor sequences. Reads were mapped to the dm6 genome using Bowtie (version.1.3.0, settings: -f -v 3 -a --best --strata --sam), allowing up to three mismatches. Genome unique reads were mapped to 1-kb tiles, normalized to library depth, and a pseudocount of '1' was added before enrichment values over input were determined. Each ChIP-seq sample was adjusted using a correction factor based on median input levels and median background levels to reach median background enrichment of 1 to correct for unequal ChIP efficiency. Replicates were averaged for genomic 1-kb tile analyses.

ChIP-seq analysis on transposon consensus sequences

Genome mapping reads longer than 23 nucleotides were mapped to TE consensus sequences using bowtie (v.1.3.0; settings: -f -v 3 -a --best --strata --sam) allowing up to 3 mismatches. Reads mapping to multiple elements were assigned to the position with the best mapping. Reads mapping to multiple positions were randomly distributed. To obtain one value per element, library depth-normalized ChIP and input reads were averaged over all nucleotide positions of each element. ChIP-seq enrichment was calculated with a pseudo count of 1 and adjusted using sample-specific correction factors determined from background 1 kb tiles to achieve median background enrichments of 1.

Small RNA-Seq Analysis

Raw reads were trimmed for linker sequences, barcodes and the 4/6 random nucleotides before mapping to the *Drosophila melanogaster* genome (dm6), using Bowtie (version.1.3.0, settings: -f -v 3 -a --best --strata --sam) with 0 mismatches allowed. Genome mapping reads were intersected with Flybase genome annotations (r6.40) using BEDTools to allow the removal of reads mapping to rRNA, tRNA, snRNA, snoRNA loci and the mitochondrial genome. For TE mappings, all genome mappers were used allowing no mismatches. Reads mapping to multiple elements were assigned to the best match. Reads mapping equally well to multiple positions were randomly distributed. Libraries were normalized to 1 million sequenced microRNA reads. For calculation of piRNAs mapping to TEs, only antisense piRNAs were considered, and counts were normalized to TE length. For classification of tiles and transposons into Rhino-independent and Rhino-dependent TEs in ovaries and testes, a binary cutoff of at a 2-fold reduction in antisense piRNA levels in *rhino* knockdown compared to control was applied based on the control samples of the respective tissue.

Local unique piRNA cluster mapping piRNAs

piRNA counts of major piRNA clusters relevant in ovaries or testes were determined using cluster definitions established by Chen et al. (Chen and Aravin, 2023 [DOI](#)). Locus-unique multi-mappers were obtained by intersecting the 5' ends of the genome aligned reads with the cluster coordinates. Only reads intersecting only with a single source locus and nowhere else in the genome were allowed. Reads mapping multiple times within one source locus were allowed but only counted once. To account for genotype differences, tiles with a read count of zero in any of the analyzed genotypes were excluded from the analysis.

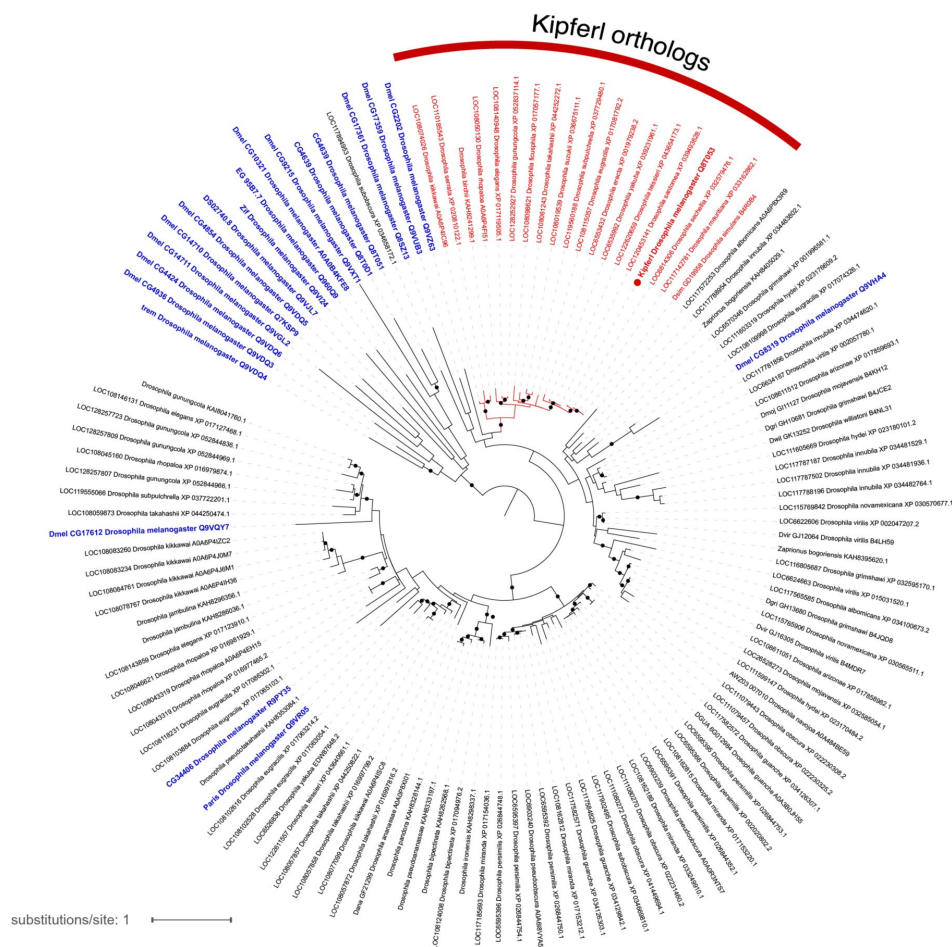


Figure 1, figure supplement 2

Phylogenetic tree illustrating the evolutionary relationship of zinc finger associated domain (ZAD)-containing zinc finger proteins based on ZAD protein sequence. Blue labels indicate *Drosophila melanogaster* proteins, red labels mark Kipferl orthologs in different species. Branches that are supported by an ultrafast bootstrap (UFBoot) value $\geq 95\%$ are indicated by a black dot. Branch lengths represent the inferred number of amino acid substitutions per site, and branch labels are composed of gene name (if available), genus, species, and accession number.

Figure 1, figure supplement 3

Diagnostic plots for rank 1-5 for the AlphaFold2 Multimer prediction of the Rhino chromodomain with the Kipferl ZnF cluster 1. (A) PAE plots (B) pLDDT plot (C) Superposition on the Rhino chromodomain of the models for rank 1 – 5, as Ca trace.

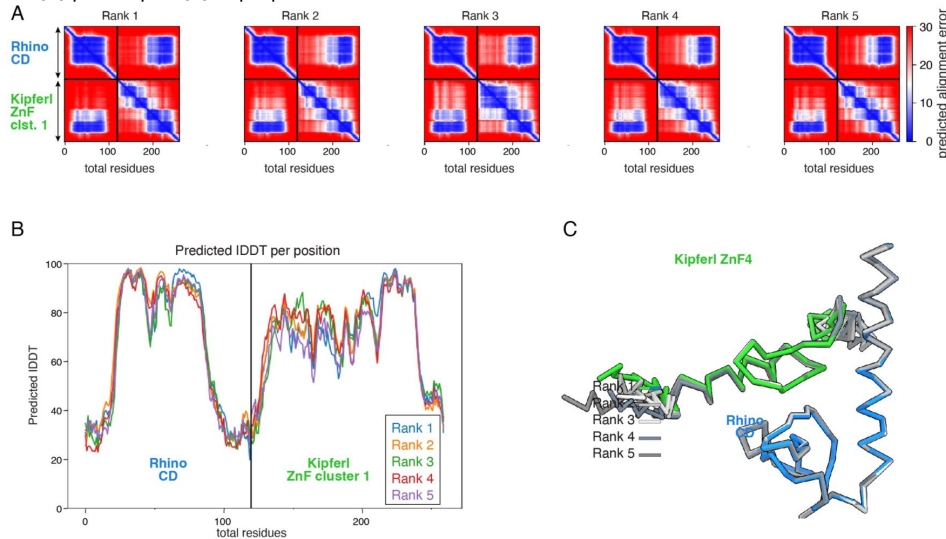


Figure 1, figure supplement 4

Multiple sequence alignment of Kipferl proteins across *Drosophila* species. Buried surface area score in blue (Krissinel and Henrick, 2007) indicates residues involved in contacts with Rhino based on the models predicted by AlphaFold. Labelled residues correspond to those highlighted in Figure 1C.

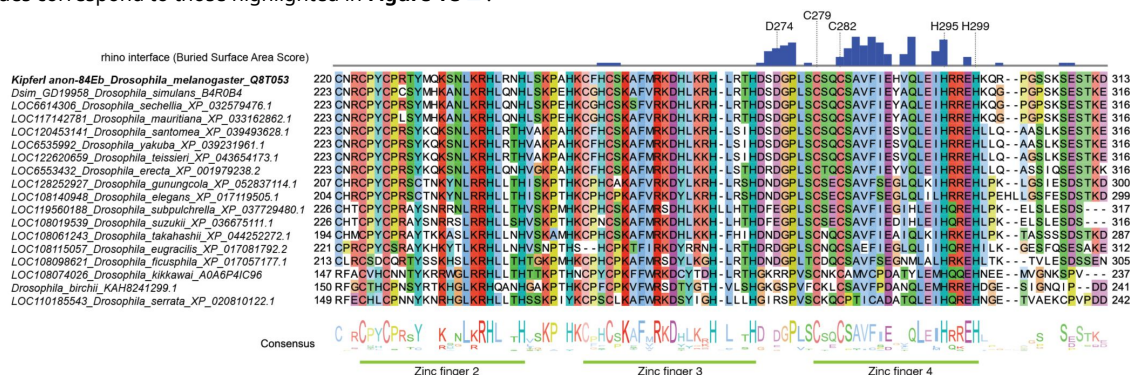
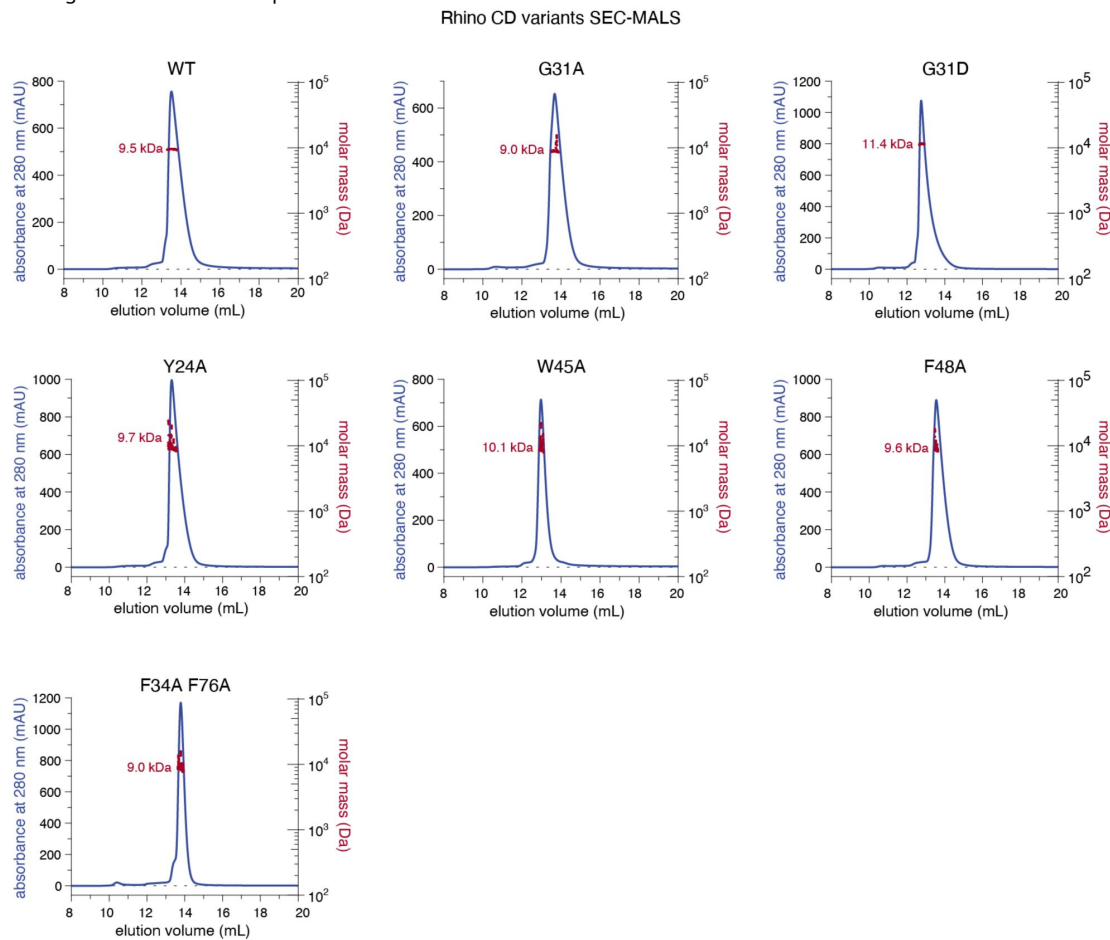


Figure 2, figure supplement 1

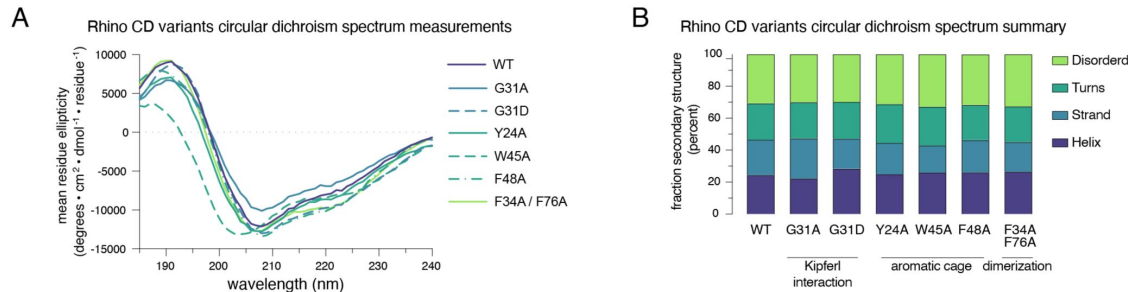
Individual line graphs depicting SEC-MALS results for the examined Rhino chromodomain constructs with in solution molecular weight measurements depicted in red.



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Figure 2, figure supplement 2

(A) Raw circular dichroism spectrum plots comparing the ellipticity versus wavelength for tested Rhino chromodomain constructs. (B) Bar graph summarizing circular dichroism spectrum measurements for the tested Rhino chromodomain constructs.



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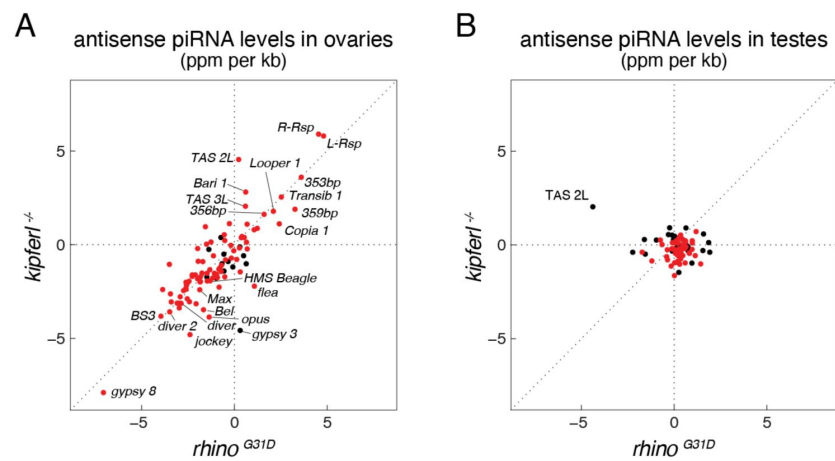
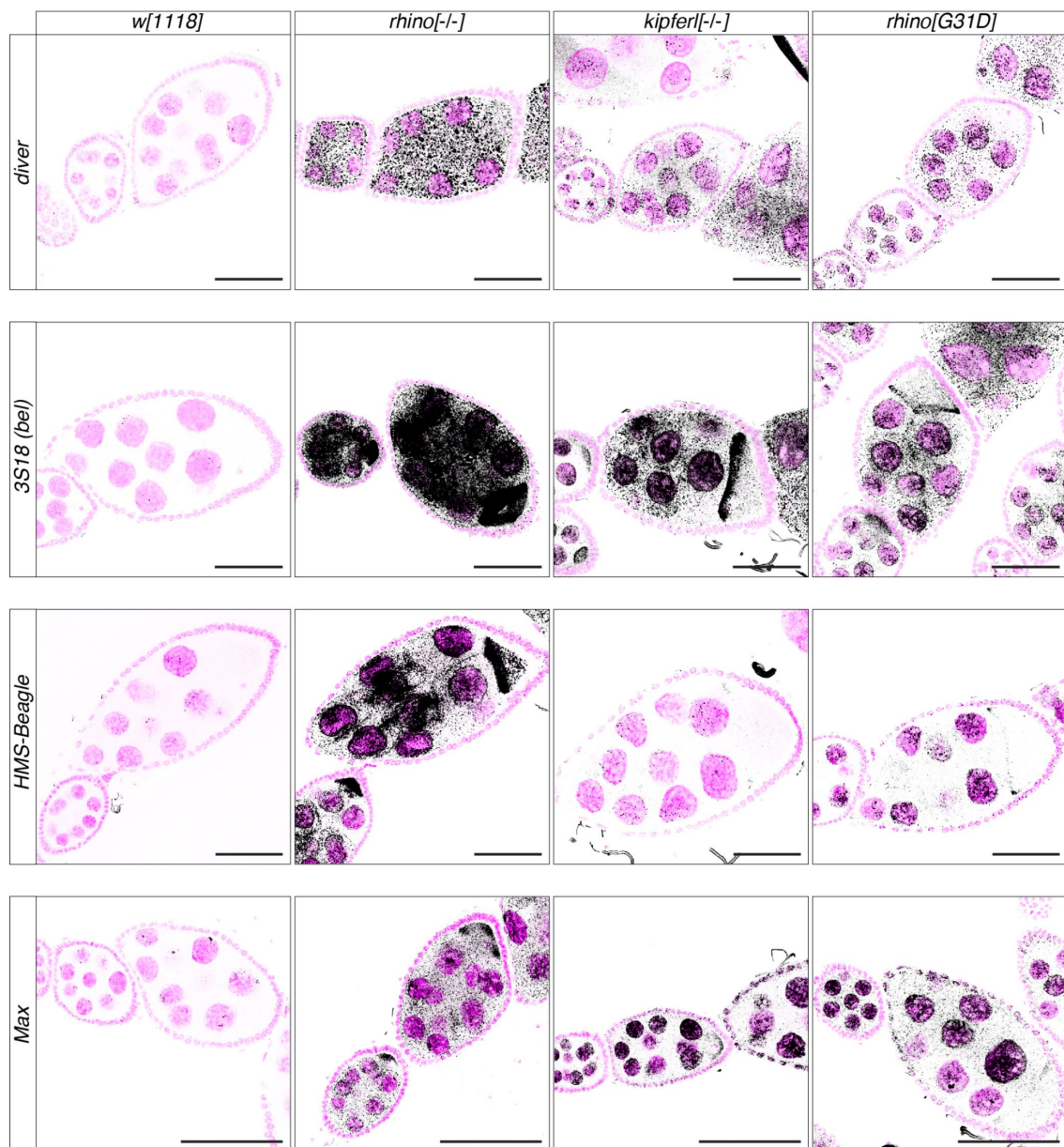


Figure 5, figure supplement 1

Scatter plots depicting the levels of antisense piRNAs mapping to transposon consensus sequences in ovaries **(A)** and testes **(B)**. Rhino-dependent elements are indicated in red.



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Figure 5, figure supplement 2

Confocal images showing RNA FISH signal (black) for transcripts of indicated transposons in *w¹¹¹⁸*, *kipferl* or *rhino* null mutant, as well as *rhino^{G31D}* ovaries. DAPI signal is displayed in pink (scale bars: 50 μ m).

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Reviewer #1 (Public Review):

Summary:

The manuscript focuses on an unexpected finding that a tiny change in a protein's amino acid sequence can redefine its biological function. The authors' data and analyses explain how a chromodomain, typically implicated in interactions with histones, can also mediate binding of HP1 homolog Rhino to the non-histone partner protein Kipferl. They elegantly pinpoint the

capacity for such interaction to a single aminoacid substitution (in fact, a single-nucleotide! substitution).

Strengths:

Both genetic and biochemical approaches are applied to rigorously probe the proposed explanation. The authors find their predictions to be borne out both in vivo, in mutant animals, and in biochemical experiments. The manuscript also features phylogenetic comparisons that put the finding into a broader evolutionary perspective.

Weaknesses pointed out in the original submission were addressed in the revised manuscript.

<https://doi.org/10.7554/eLife.93194.2.sa1>

Reviewer #3 (Public Review):

Summary:

This article is a direct follow-up to the paper published last year in eLife by the same group. In the previous article, the authors discovered a zinc finger protein, Kipferl, capable of guiding the HP1 protein Rhino towards certain genomic regions enriched in GRGGN motifs and packaged in heterochromatin marked by H3K9me3. Unlike other HP1 proteins, Rhino recruitment activates the transcription of heterochromatic regions, which are then converted into piRNA source loci. The molecular mechanism by which Kipferl interacts specifically with Rhino (via its chromodomain) and not with other HP1 proteins remained enigmatic.

In this latest article, the authors go a step further by elucidating the molecular mechanisms important for the specific interaction of Rhino and not other HP1 proteins with Kipferl. A phylogenetic study carried out between the HP1 proteins of 5 *Drosophila* species led them to study the importance of an AA Glycine at position 31 located in the Rhino chromodomain, an AA different from the AA (aspartic acid) found at the same position in the other HP1 proteins. The authors then demonstrate, through a series of structure predictions, biochemical and genetic experiments, that this specific AA in the Rhino-specific chromodomain explains the difference in the chromatin binding pattern between Rhino and the other *Drosophila* HP1 proteins. Importantly, the G31D conversion of the Rhino protein prevents interaction between Rhino and Kipferl, phenocopying a Kipfer mutant.

Strengths:

The strength of this study is to test at the molecular and genetic level whether the difference in the AA sequence- encouered by phylogenetic analysis of HP1 proteins including Rhino combined with structure prediction- can explain the difference in chromatin binding patterns between HP1 proteins and Rhino.

To do so they have created a Rhino mutant by introducing a point mutation into the endogenous rhino gene, reverting the Glycine in position 31 to the aspartic acid found in all other HP1 proteins. Even if the Rhino G31D mutant retains its ability to interact with H3K9me3 (predictive and biochemistry approaches that I'm less familiar with) it does not localize correctly on the chromatin preventing certain regions such as locus 80F from being converted into piRNA source loci. However other regions such as satellite regions attract the Rhino mutant protein converting them into super piRNA source loci, phenocopying the effects observed in a Kipferl mutant. Why Rhino when not bound to Kipferl concentrates in satellite regions is a question that remains unanswered.

Weaknesses:

In this new version of the manuscript, the authors have answered all the questions and weaknesses raised previously.

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Author response:

The following is the authors' response to the original reviews.

Public Review:

This article is a direct follow-up to the paper published last year in eLife by the same group. In the previous article, the authors discovered a zinc finger protein, Kipferl, capable of guiding the HP1 protein Rhino towards certain genomic regions enriched in GRGGN motifs and packaged in heterochromatin marked by H3K9me3. Unlike other HP1 proteins, Rhino recruitment activates the transcription of heterochromatic regions, which are then converted into piRNA source loci. The molecular mechanism by which Kipferl interacts specifically with Rhino (via its chromodomain) and not with other HP1 proteins remained enigmatic.

In this latest article, the authors go a step further by elucidating the molecular mechanisms important for the specific interaction of Rhino and not other HP1 proteins with Kipferl. A phylogenetic study carried out between the HP1 proteins of 5 Drosophila species led them to study the importance of an AA Glycine at position 31 located in the Rhino chromodomain, an AA different from the AA (aspartic acid) found at the same position in the other HP1 proteins. The authors then demonstrate, through a series of structure predictions, biochemical, and genetic experiments, that this specific AA in the Rhino-specific chromodomain explains the difference in the chromatin binding pattern between Rhino and the other Drosophila HP1 proteins. Importantly, the G31D conversion of the Rhino protein prevents interaction between Rhino and Kipferl, phenocopying a Kipferl mutant.

Strengths:

The authors' effective use of phylogenetic analyses and protein structure predictions to identify a substitution in the chromodomain that allows Rhino's specific interaction with Kipferl is very elegant. Both genetic and biochemical approaches are applied to rigorously probe the proposed explanation. They used a point mutation in the endogenous locus that replaces the Rhino-specific residue with the aspartic acid residue present in all other HP1 family members. This novel allele largely phenocopies the defects in hatch rate, chromatin organization, and piRNA production associated with kipferl mutants, and does not support Kipferl localization to clusters. The data are of high quality, the presentation is clear and concise, and the conclusions are generally well-supported.

Weaknesses:

The reviewers identified potential ways to further strengthen the manuscript.

(1) The one significant omission is RNAseq on the rhino point mutant, which would allow direct comparison to cluster, transposon, and repeat expression in kipferl mutants.

In this eLife Advances submission, we aim to elucidate the molecular interaction between Rhino and the zinc finger protein Kipferl and how it evolved. Using various assays, of which piRNA sequencing is the most relevant and comprehensive, we show that the rhino[G31D]

mutation phenocopies a rhino loss-of-function situation for Kipferl and a kipferl loss-of-function situation for Rhino. Further confirmation of this statement by additional RNA-seq experiments to probe the extent of selective TE de-repression would indeed be a possibility. We decided to test for TE de-repression phenotypes using sensitive RNA-FISH experiments of a handful of TEs that are deregulated in kipferl loss of function flies (Baumgartner et al. 2022). This showed that the same TEs are also deregulated in rhino[G31D] flies, further confirming the similarity of the two genotypes. We have added these data to the text and to Figure 5-figure supplement 2, which shows representative RNA FISH images.

(2) The manuscript would benefit from adding more evolutionary comparisons. The following or similar analyses would help put the finding into a broader evolutionary perspective:

i) Is Kipferl's surface interacting with Rhino also conserved in Kipferl orthologs? In other words, are the Rhino-interacting amino acids of Kipferl under any pressure to be conserved?

We performed an analysis of the Kipferl interface that interacts with the Rhino chromodomain in those species where Kipferl could be unambiguously identified. This showed that the residues involved in the Rhino interaction are generally conserved. We have added this analysis to Figure 1-figure supplement 4.

ii) The remarkable conservation of Rhino's G31 is at odds with the arms race that is proposed to be happening between the fly's piRNA pathway proteins and transposons. Does this mean that Rhino's chromodomain is "untouchable" for such positive selection?

We agree that the conservation of the G31 residue argues against this binding interface being under positive selection in Rhino. Without understanding the pressures acting on Rhino that underlie the previously published positive selection, we find it difficult to draw firm conclusions. Mutating G31 in fly species that lack Kipferl would be an interesting experiment.

Recommendations for the authors:

(1) RNAseq is important to the full characterization of the phenotype and should be included. It's now clear that the major piRNA clusters are not required for fertility, so I would also include an analysis of piRNA production and Rhino binding to regions flanking isolated insertions.

See our response to raised weakness #1 above. Briefly, we have now added an analysis of TE de-repression based on RNA-FISH experiments (Figure 5-figure supplement 2). Regarding the proposed analysis of piRNA production and Rhino binding to regions flanking isolated TE insertions: this is an important issue that we carefully analysed in our previous work characterising the kipferl mutant (Baumgartner et al. 2022). In the present work, we focused on generating a rhino mutant that uncouples Rhino from Kipferl.

(2) The authors do not provide direct biochemical evidence that the chromodomain substitution blocks Rhino binding to Kipferl. However, Rhino protein is very low abundance, making analysis of the endogenous protein very difficult.

Based on our previous work (Baumgartner et al 2022), the Rhino chromodomain interacts directly with the fourth zinc finger of Kipferl. Mutation of a single residue in the predicted interface (Rhino[G31D]) phenocopies a kipferl mutant, strongly suggesting that this mutation disrupts the Rhino-Kipferl interaction. Definitive evidence will have to await the reconstitution of this interaction using recombinant proteins. Our attempts to purify recombinant Kipferl (expressed in bacteria or in insect cells) or the protein fragments

relevant to the interaction were unsuccessful so far. While we obtained soluble fractions of the first ZnF array, there was always a high level of co-purifying nucleic acids that we were not able to remove.

(3) Even if the Rhino G31D mutant retains its ability to interact with H3K9me3 it does not localize correctly on the chromatin preventing certain regions such as locus 80F from being converted into piRNA source loci. However other regions such as satellite regions attract the Rhino mutant protein converting them into super piRNA source loci, phenocopying the effects observed in a Kipferl mutant. Why Rhino when not bound to Kipferl concentrates in satellite regions is a question that remains unanswered.

This is a very interesting question indeed. We have not been able to elucidate the molecular basis of how Rhino is recruited to satellite repeats in Kipferl mutants. For example, we performed a proximity biotinylation experiment with GFP-Rhino in Kipferl mutant ovaries, but this experiment did not reveal any protein that would explain the observed accumulation of Rhino at the complex satellite repeats.

(4) In the phylogenetic analysis the authors identified two residues as Rhino-specific and conserved sequence alterations, the D31G mutation and the G62 insertion. However, the authors limit their study to D31G mutation, and nothing is performed on the G32 insertion. It would have been interesting to know the impact of this insertion on Rhino's biology.

The role, if any, of the Rhino-specific G62 insertion and its effect on Rhino localisation or function is an interesting topic for further study. We have not investigated the G62 residue experimentally. In the current manuscript, we limited our efforts to the analysis of the G31D mutation, as the goal was to identify the mode of interaction with Kipferl, and the G62 residue is not predicted to contact Kipferl according to AlphaFold.

(5) The authors report that the G31D mutation of Rhino phenocopies the Kipferl mutant. Rhino is wrongly localized in the nucleus, and Rhino G31D recruitment in certain Kipferl-enriched regions is affected, as at the 80F locus, which correlates with a strong drop in piRNA production from this locus. To go a step further in demonstrating that G31D phenocopies the Kipferl mutant, it would have been informative to analyse how much TE piRNAs are affected and whether TEs are deregulated.

See our response to similar comments above. We have added RNA-FISH experiments to illustrate that the TE de-repression phenotypes are comparable between rhino[G31D] and kipferl loss of function ovaries (Figure 5-figure supplement 2). Analyses of TE-mapping piRNAs also show well correlated phenotypes (Figure 5-figure supplement 1).

(6) Figure 3A: To homogenize with the immunostaining presented in Figure 3B, can the authors add on the bar graph depicting female fertility the results obtained with kipferl^{-/-} and rhino^{-/-} genotype?

rhino mutants are completely (100%) sterile and the fertility of kipferl mutants was previously measured to range between 15% and 40% (Baumgartner et al. 2022).

(7) Figure 4A: It would have been interesting to show Venn diagrams showing the overlap of genomic regions enriched for Kipferl versus regions enriched for Rhi in a WT and in a Rhi G31D mutant.

We consider the analysis presented in Figure 4 to be more meaningful, as a Venn diagram would require binary cut-offs.

(8) *Figure 1B: In the phylogenic analysis for Rhino/HP1d two D. simulans lines are presented. Can the authors clarify this point?*

There are two Rhino paralogs in *D. simulans*: one paralog (NCBI: AAY34025.1) is more similar to *D. melanogaster* Rhino, contains one intron and is located at chromosome chr2R (assembly Apr. 2005, WUGSC mosaic 1.0/droSim1: 12256895-12258668). The second paralog (XP_002106478.1) is located on chromosome X (6734493-6735248) and does not contain an intron. We have added a clarifying statement to the corresponding figure legend.

(9) *To determine whether Rhino G31D point mutation affects the overall function of Rhino, the authors analysed Kipferl-independent piRNA source loci by looking at Responder and 1,688 family satellites. I'm not sure that these loci can be classified as Kipferl-independent piRNA source loci since a strong increase of piRNA production from these loci in Kipferl mutant is observed. In my point of view, the 42AB and 38C are real Kipferl-independent piRNA source loci as piRNA production from these loci is not affected by Kipferl KD.*

Indeed, the Rsp and 1,688 family satellites are not completely independent of Kipferl, as their expression and Rhino occupancy differ between wild-type and kipferl loss-of-function phenotypes (including rhino[G31D]). However, we believe that this increase is due to a strong dependence on different sequestration mechanisms and is not mediated by a direct function of Kipferl at these sites. Similarly, we observe slight differences in piRNA production for the peripheral parts of cluster 42AB, as well as differences in Rhino occupancy despite an unaltered piRNA profile at cluster 38C (Baumgartner et al. 2022). Thus, different flavours of Kipferl-independence exist, with the only truly Kipferl-independent piRNA sources likely to be the piRNA clusters in the testis. A clear classification is further complicated by previously observed compensatory effects in the piRNA pathway, leading us to adopt the current definition of "requiring Kipferl for Rhino recruitment" to distinguish Kipferl-dependent from Kipferl-independent sites.

(10) *The authors report that the G31D mutation of Rhino phenocopies the Kipferl mutant. Rhino is wrongly localized in the nucleus, and Rhino G31D recruitment in certain Kipferl-enriched regions is affected, as at 80F locus, which correlates with a strong drop in piRNA production from this locus. To go a step further in demonstrating that G31D phenocopies the Kipferl mutant, it would have been interesting to look at how much TE piRNAs are affected and whether TEs (and which class of TE) are deregulated by RNAseq and/or in situ hybridization.*

See our response to similar comments above. Our new RNA-FISH experiments and TE-mapping piRNA analysis extend the comparison of phenotypes between kipferl mutants and rhino[G31D] mutants and are consistent with our previous conclusions (Figure 5-figure supplements 1 and 2).

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